

Deep Learning Neural Networks 2 – Exam Questions

Students Edition

26.6.2020

1 Questions

1.1 Question 1

Which of the following statements about AlexNet (Krizhevsky et al., 2012) are correct?

Wählen Sie eine oder mehrere Antworten:

- a. AlexNet uses input images with a resolution of 224x224 pixels and three channels.
- b. AlexNet is a deep convolutional network with more than 1 hidden layer.
- c. AlexNet does not have fully-connected layers.
- d. One of the most important components of AlexNet is BatchNorm.
- e. AlexNet uses strided convolutions instead of max-pooling for down-sampling.

1.1.1 Answers

Correct answers are a. and b.

1.2 Question 2

Which of the following statements about Residual Networks (ResNets) are correct?

Wählen Sie eine oder mehrere Antworten:

- a. Residual networks use skip-connections which add the representation $\mathbf{h}^l$ of a former layer l to a later layer $l + 1$, that is: A

$$\mathbf{h}^{l+1} = \mathbf{h}^l + F(\mathbf{h}^l)$$

where F can be any transformation in a neural network with appropriate dimensions, such as a fully-connected layer.



Figure 1: Question 3. Road Scenes.

- b. Because of skip-connections, very deep Residual Networks can be trained even without normalization techniques.
- c. Leaky ReLUs cannot be used in Residual Networks.
- d. Residual networks are always convolutional networks and cannot be used in networks that have only fully-connected layers.
- e. Pre-trained Residual Networks cannot be used for transfer learning because the learned features are usually overly specified to the task on which they were trained.

1.2.1 Answers

Correct answer is a.

1.3 Question 3

Imagine you are working on a machine learning project related to self-driving cars.

Figure 1 shows some road scenes.

A camera inside a car provides images of road scenes such as the ones in the picture above (left column). Your task is to train a neural network that supplies images on as shown on the right hand side of the picture (right column),

in which cars, roads, pedestrians, etc are marked as shown. Which of the following statements are true in this context?

Wählen Sie eine oder mehrere Antworten:

- a. Generative adversarial networks are the most appropriate type of networks to approach this task.
- b. The intersection over union (IoU) would not be an appropriate metric to assess the performance of model is trained on this task.
- c. This represents an object detection task in computer vision.
- d. In order to train such a network, labels for each pixel of the training set images are required.
- e. Deep Neural networks cannot be used for this task because there is no appropriate loss function.

1.3.1 Answers

Correct answer is d.

1.4 Question 4

Assume you are given a convolutional neural network with a fixed architecture. You are only allowed to change the parameters of last convolutional layer and you want to strongly increase the receptive field (RF) of each neuron in the last layer.

For each of the following four possible changes "increasing kernel size", "increase number of kernels", "increase stride", or "switch to dilated convolutions" decide whether they strongly increase, slightly increase or do not increase the receptive field (RF) of your network!

- Switch to dilated convolutions
- Increase stride of convolution
- increase the number of kernels
- increase kernel size

1.4.1 Answers

Correct answers are:

- Switch to dilated convolutions → strong increase of RF,
- Increase stride of convolution → strong increase of RF,

- increase the number of kernels $\rightarrow$ no increase of RF,
- increase kernel size $\rightarrow$ small increase of RF

1.5 Question 5

You are given an input $\mathbf{x} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ and a kernel weight matrix $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$.

What is the result of the transpose convolution operation $*^T$ (stride 1) of

$$\mathbf{w} *^T \mathbf{x} = ?$$

wählen Sie eine Antwort:

$$a = \begin{pmatrix} 0 & 2 & 1 \\ 6 & 14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$$

$$b. = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{pmatrix}$$

$$c. = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 0 & 3 \\ 4 & 6 & 6 & 9 \end{pmatrix}$$

$$d = \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$$

$$e. = \begin{pmatrix} 13 & 13 & 13 \\ 13 & 13 & 13 \\ 13 & 13 & 13 \end{pmatrix}$$

$$f. = (42)$$

$$g = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix}$$

$$h. = (14)$$

1.5.1 Answers

Correct answer is:

$$g = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix}$$

1.6 Question 6

Which of the following models or architectures are one-stage and which are two-stage approaches?

- Mask R-CNN
- Faster R-CNN
- R-CNN
- Fast R-CNN
- YOLO
- SSD

1.6.1 Answers

Correct answer is:

- Mask R-CNN → Two-stage,
- Faster R -CNN → Two-stage,
- R -CNN → Two-stage,
- Fast R -CNN → Two-stage,
- YOLO → One-stage,
- SSD → One-stage

1.7 Question 7

Which of the following statements about DeepLearning-based object detection methods are correct?

Wählen Sie eine oder mehrere Antworten:

- Fast R-CNN has two output layers (also named heads): one for regression and one for classification.
- The operations ROI-pooling and ROI align operate on feature maps and not on the input image.
- Object detection can be formulated as regression task in a multi-task setting.
- R-CNN uses AlexNet to extract features for Regions Of Interest.
- Faster R-CNN uses a region proposal network rather than the selective search algorithm that was used in R-CNN.

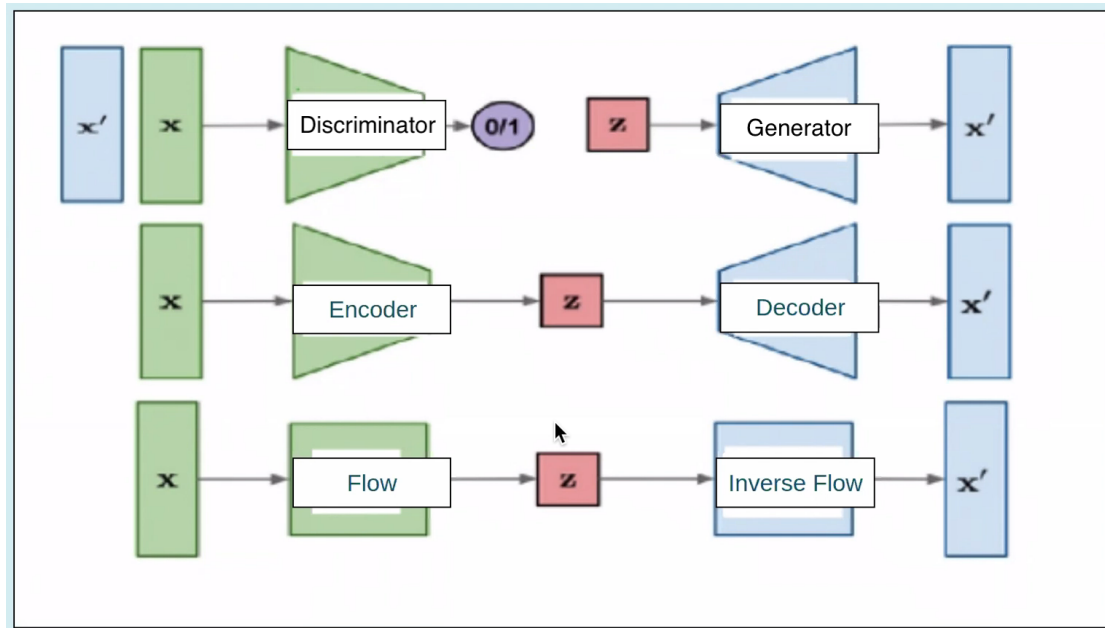


Figure 2: Question 8. Models.

1.7.1 Answers

Correct answers are (all above):

- R-CNN uses AlexNet to extract features for Regions Of Interest.,
- Object detection can be formulated as regression task in a multi-task setting.,
- Fast R-CNN has two output layers (also named heads): one for regression and one for classification.,
- Faster R-CNN uses a region proposal network rather than the selective search algorithm that was used in R-CNN.,
- The operations ROI-pooling and ROI align operate on feature maps and not on the input image.

1.8 Question 8

In this picture you can see three categories of generative models we discussed in the lecture. Annotate the boxes in the figure with the following terms, so that the main building blocks/networks are assigned correctly: "Generator", "Discriminator", "Encoder", "Decoder", "Flow", "Inverse Flow". Move the boxes on the

bottom in the correct way into the grey boxes in the main figure. To be clear, a rectangular shape of the network means that input and output dimensions stay the same, whereas a trapezoidal shape means decreasing (or increasing) dimension in the respective directions. Figure 2 shows some models.

1.8.1 Answers

Correct answer is:

Figure 2 shows the correct answers.

1.9 Question 9

Which of the following statements about autoencoders (not including variational autoencoders) are correct?

Wählen Sie eine oder mehrere Antworten:

- a. Autoencoders are always fully-connected networks.
- b. Autoencoders do not map a single input data point to a single point in latent space, but they map a single input data point to a distribution in latent space.
- c. The Input-Output-Jacobian of an autoencoder is always a lower-triangular matrix.
- d. Autoencoders consist of an encoder and a decoder, which are parametrized neural networks.
- e. All autoencoders use fewer neurons in the hidden layers than in the input layer.

1.9.1 Answers

Correct answer is: d.

1.10 Question 10

Assume you are given two univariate Gaussian distributions. The first one, p_1 has mean zero and unit variance ($\mathcal{N}(0, 1)$). The second Gaussian p_2 has also unit variance, but a mean at $x > 0$, hence: $\mathcal{N}(x, 1)$. You are interested in the Kullback-Leibler divergence $\text{KL}(p_1 \| p_2)$, the Jensen-Shannon Divergence $\text{JSD}(p_1 \| p_2)$, and the Wasserstein-2 distance $\text{WD}(p_1 \| p_2)$ between these two distributions. You now increase x and plot this value against the divergence or distance measures, which gives three graphs displayed in the figure below. Which divergence belongs to which graph?

Figure 3 for reference.

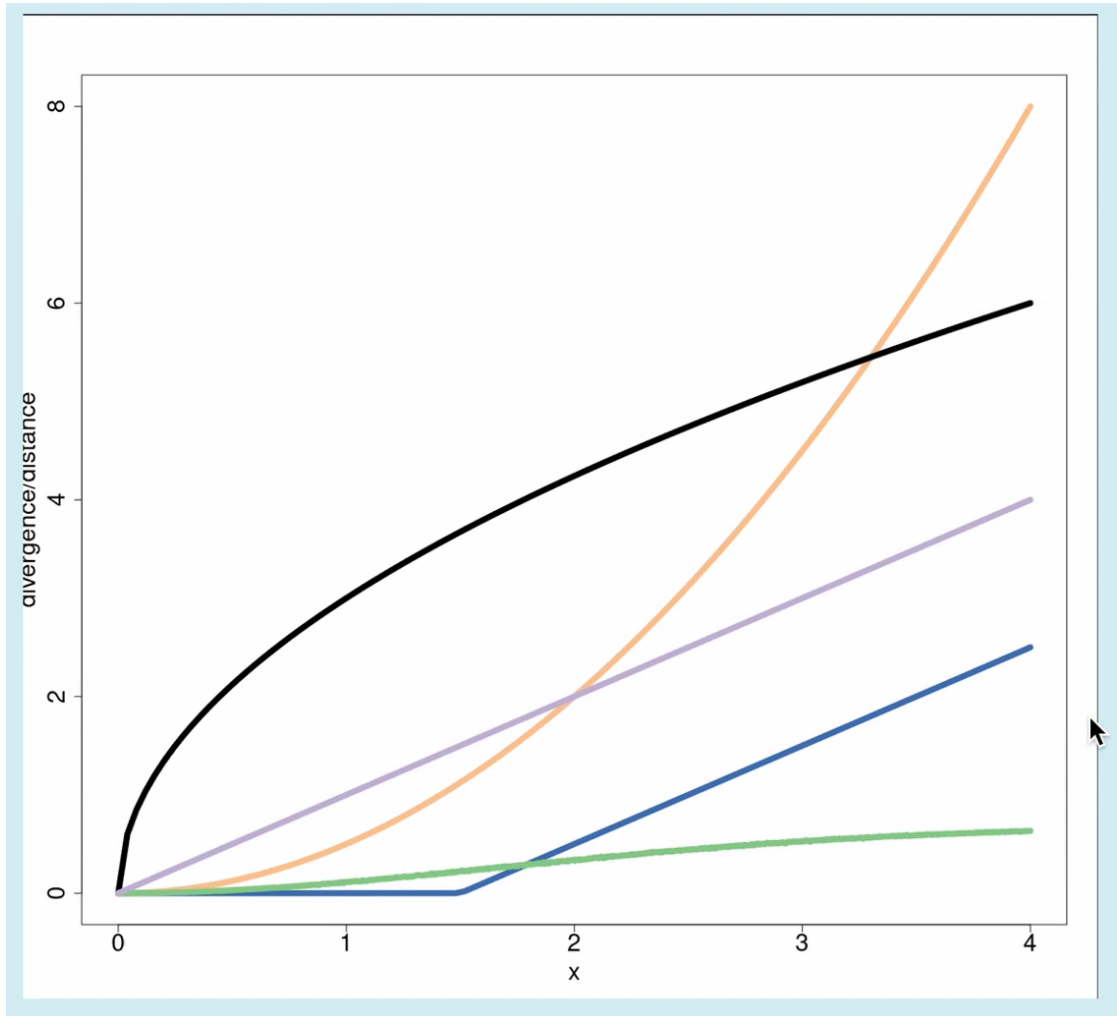


Figure 3: Question 10. Divergences and Distances of distributions.

1.10.1 Answers

Correct answer is:

- Wasserstein distance: → Purple line,
- Jensen-Shannon divergence: → Green line,
- KL divergence: → Orange line

1.11 Question 11

Using the notation introduced in the lecture, which of the following loss functions are correct formulations of the standard Variational Autoencoder (VAE) loss function?

Wählen Sie eine Antwort:

- a. $\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} | \mathbf{z}) - D_{KL}(q_\phi(\mathbf{z} | \mathbf{x}) || p_\theta(\mathbf{z}))$
- b. $\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} | \mathbf{z}) - D_{KL}(p_\phi(\mathbf{x}) || p_\theta(\mathbf{z}))$
- c. $\mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [D(G(\mathbf{z}))]$
- d. $\mathcal{L}_\theta(\mathbf{x}) = D_{KL}(p_\mathbf{x}(\mathbf{x}; \boldsymbol{\theta}) || p_\mathbf{x}^*(\mathbf{x}))$
- e. $\mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))]$

1.11.1 Answers

Correct answer is: a. $\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} | \mathbf{z}) - D_{KL}(q_\phi(\mathbf{z} | \mathbf{x}) || p_\theta(\mathbf{z}))$

1.12 Question 12

Which of the following statements about variational autoencoders (VAEs) are true?

Wählen Sie eine oder mehrere Antworten:

- a. Variational autoencoders cannot be used to generate images.
- b. The prior distribution of the variables $\mathbf{z}$ in the latent space must be a discrete distribution for VAEs.
- c. Variational autoencoders have an encoder-decoder structure, where the encoder and decoder must be symmetric.
- d. Variational autoencoders maximize a lower bound on the likelihood of the data.
- e. Variational autoencoders automatically provide the marginal likelihood of a data point $p(\mathbf{x})$ at low computational costs.

1.12.1 Answers

Correct answer is: d. Variational autoencoders maximize a lower bound on the likelihood of the data.

1.13 Question 13

Using the notation introduced in the lecture, which of the following objective functions are correct formulations of the standard objective function of Generative Adversarial Networks (GANs)?

Wählen Sie eine Antwort:

- a. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(D(G(\mathbf{z})))]$
- b. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))]$
- c. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log G(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - G(D(\mathbf{z})))]$
- d. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_r} [\log(1 - D(G(\mathbf{x})))]$
- e. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [D(G(\mathbf{z}))]$
- f. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(\mathbf{z}))]$

1.13.1 Answers

Correct answer is: b. $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))]$

1.14 Question 14

Consider a Generative Adversarial Network (GAN) which successfully produces images of apples. Which of the following statements are correct?

Wählen Sie eine oder mehrere Antworten:

- a. After training the GAN, the discriminator loss eventually reaches a constant value.
- b. The discriminator can be used to classify images as apple vs. non-apple.
- c. The generator can produce unseen images of apples.
- d. The generator aims to learn the distribution of apple images.

1.14.1 Answers

Correct answers are: a., c. and d.

The generator aims to learn the distribution of apple images.,

After training the GAN, the discriminator loss eventually reaches a constant value.,

The generator can produce unseen images of apples.

1.15 Question 15

Tick the correct statements related to the Inception Score (IS) and the Frechet Inception Distance (FID).

Wählen Sie eine oder mehrere Antworten:

- a. IS and FID are metrics that assess the quality of GAN-generated images.
- b. The computation of the FID is based on an explicit formula for the Wasserstein 2 -distance between two Gauss distributions.
- c. Both IS and FID rely on features for input images that are produced by AlexNet.
- d. The FID in general is better suited to detect mode collapse than the IS, since it takes into account the covariances.
- e. IS and FID can be used as GAN objectives that lead to a faster training procedure.

1.15.1 Answers

Correct answers are: a., b., and d.

The computation of the FID is based on an explicit formula for the Wasserstein 2 -distance between two Gauss distributions.,

IS and FID are metrics that assess the quality of GAN-generated images.,

The FID in general is better suited to detect mode collapse than the IS, since it takes into account the covariances.

1.16 Question 16

Decide for each of the statements below, if they are more likely for a GAN or a WGAN.

- The discriminator performs regression.
- There are log-terms in the objective.
- The objective is likely to suffer from the vanishing gradient problem.
- Uses many more optimization steps for the discriminator than for the generator.
- No weight-clipping is needed.

1.16.1 Answers

Correct answers are:

- The discriminator performs regression. → WGAN,
- There are log-terms in the objective. → GAN,

- The objective is likely to suffer from the vanishing gradient problem. $\rightarrow$ GAN,
- Uses many more optimization steps for the discriminator than for the generator. $\rightarrow$ WGAN,
- No weight-clipping is needed. $\rightarrow$ GAN

1.17 Question 17

Assume that you are given a two-dimensional vector $\mathbf{u} \sim p_{\mathbf{u}}(\mathbf{u})$. Moreover, let $T = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{x} = T \cdot \mathbf{u}$. Mark the correct formula for the distribution $p_{\mathbf{x}}(\mathbf{x})$

Wählen Sie eine Antwort:

- a. $p_{\mathbf{u}} \left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot \frac{1}{2}$
- b. $p_{\mathbf{u}} \left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot 2$
- c. $p_{\mathbf{u}} \left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot 2$
- d. $p_{\mathbf{u}} \left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot \frac{1}{2}$
- e. $p_{\mathbf{u}} \left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right)$
- f. $p_{\mathbf{u}} \left(\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot \frac{1}{2}$

1.17.1 Answers

Correct answer is: a. $p_{\mathbf{u}} \left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x} \right) \cdot \frac{1}{2}$

1.18 Question 18

Which of the following statements related to flow-based models are correct?

Wählen Sie eine oder mehrere Antworten:

- a. For Autoregressive flows the involved transformation g_{ϕ} is constructed in a way such that its Jacobi determinant always has value 1.
- b. The computation of the Jacobi determinant of a neural network with D inputs and D outputs is in general of $\mathcal{O}(D^3)$
- c. Fitting flow based models is often done with the KL-divergence.

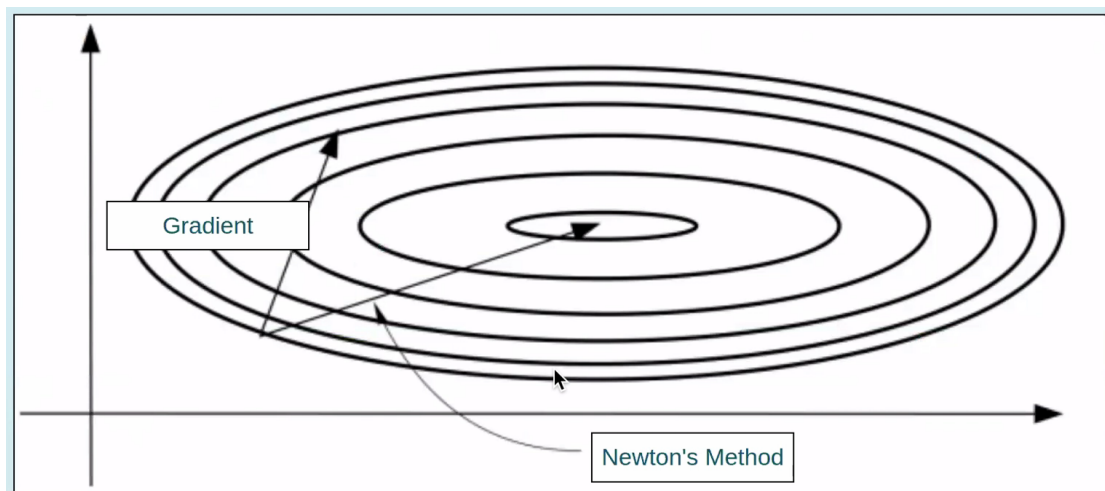


Figure 4: Question 19. Newtons Method.

d. Flow based models can represent any distribution $p_{\mathbf{x}}(\mathbf{x}) > 0$ if one starts from a sufficiently complex distribution. A simple base distribution $p_{\mathbf{u}}(\mathbf{u})$ like the uniform distribution on the cube $(0, 1)^D$ would not be sufficient.

1.18.1 Answers

Correct answers are: b. and c.

Fitting flow based models is often done with the KL-divergence.,

The computation of the Jacobi determinant of a neural network with D inputs and D outputs is in general of $\mathcal{O}(D^3)$. .

1.19 Question 19

In this picture you can see the update direction for Newton's method and the gradient direction for a quadratic error surface (i.e. the level curves origin from a quadratic form like an ellipsis). Annotate the corresponding directions/arrows in the figure with the following terms: "Gradient", "Newton's Method". Move the boxes on the bottom in the correct way into the grey boxes in the main figure.

Figure 4 for reference.

1.19.1 Answers

Correct answer is: Figure 4 shows the correct answers.

1.20 Question 20

Mark the correct statements related to the Conjugated gradient (CG) method.

Wählen Sie eine oder mehrere Antworten:

- a. The reference matrix establishes a link between updates with respect to conjugate directions and optimization.
- b. Conjugated vectors are always linearly independent.
- c. In contrast to CGs, the Levenberg-Marquardt algorithm is a trust-region method.
- d. The CG-method computes conjugate vectors from previous updates and gradients at the current point.
- e. Hestenes-Stiefel and Pollak-Ribière are methods used to estimate the eigenvalues of the Hessian. x .
- f. Conjugated vectors are always orthogonal to each other. x
- g. A set of nonzero vectors, $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, is said to be conjugated with respect to some symmetric positive definite matrix $\mathbf{A}$, if $\mathbf{x}_i^T \mathbf{A} \mathbf{x}_j < 0$ for all $i \neq j$

1.20.1 Answers

Correct answers are: a., b., c. and d.

Conjugated vectors are always linearly independent.,

The reference matrix establishes a link between updates with respect to conjugate directions and optimization.,

The CG-method computes conjugate vectors from previous updates and gradients at the current point.,

In contrast to CGs, the Levenberg-Marquardt algorithm is a trust-region method.

1.21 Question 21

Mark the correct statements related to Hessian approximation and Newton or Quasi-Newton methods.

Wählen Sie eine oder mehrere Antworten:

- a. The BFGS algorithm uses a single outer product as Hessian approximation in each iteration step.
- b. Diagonal approximation is a very efficient approximation method for the Hessian, however it is not very precise and may lead to false conclusions.
- c. Quasi-Newton methods are known to estimate the Hessian from Gradients instead of actually using it.
- d. Newton's method applied to a function f always converges to a root of f , no matter where the iteration starts.

e. The Sherman-Morrison formula yields a good approximation of the eigenvalues of the Hessian.

1.21.1 Answers

Correct answers are: b. and c.

Diagonal approximation is a very efficient approximation method for the Hessian, however it is not very precise and may lead to false conclusions.,

Quasi-Newton methods are known to estimate the Hessian from Gradients instead of actually using it.

1.22 Question 22

Mark the correct statments related to the Universal Approximator Theorem (UAT) for Neural Networks (NNs).

Wählen Sie eine oder mehrere Antworten:

a. The UAT gives a useful algorithm to compute the optimal weights of a NN that should approximate a given continuous function.

b. In the case of a real-valued function h defined on $[-\pi, \pi]^2$, the existence of an approximating NN can be derived with the help of Fourier series.

c. The algorithm related to the UAT is easy to understand and implement, it is however very costly and therefore it is hardly used in practise.

d. The UAT shows that any function can be approximated by any given NN with arbitrary precision, if the weights are chosen appropriately. This means that there are no restriction on the function to be approximated and on the structure of the network that should do the job.

e. The UAT is an abstract statement about the expressive power of NNs.

1.22.1 Answers

Correct answers are: b. and e.

The UAT is an abstract statement about the expressive power of NNs.,

In the case of a real-valued function h defined on $[-\pi, \pi]^2$, the existence of an approximating NN can be derived with the help of Fourier series.

1.23 Question 23

Mark the correct statements related to Random Matrix Theory (RMT):

Wählen Sie eine oder mehrere Antworten:

- a. The quarter circular law is a statement about the behaviour of singular values for any given random matrix, no matter how the entries are distributed, if they are statistically independent or not, etc.
- b. RMT provides information about the eigenvalues or singular values of large matrices with random variables as entries.
- c. The empirical distribution of the eigenvalues of a symmetric matrix yields the average number of eigenvalues that lie in a certain region of the real line.
- d. RMT only yields useful results when one works with square matrices. x .

1.23.1 Answers

Correct answer is: b. and c.

RMT provides information about the eigenvalues or singular values of large matrices with random variables as entries.,

The empirical distribution of the eigenvalues of a symmetric matrix yields the average number of eigenvalues that lie in a certain region of the real line.